

CERTIFICATE OF ANALYSIS

Calcium Nitrate Tetrahydrate — Industrial Grade

Product Name: Calcium Nitrate Tetrahydrate	Grade: Industrial Grade
CAS No.: 13477-34-4	EINECS EC No.: 233-332-1
Molecular Formula: $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$	Molecular Weight: 236.15 g/mol
Batch No.: HRSU/CNP/04/42/2025	Manufacturing Date: 11/04/2025
Shelf Life: 60 months (typical for this product class)	Expiry Date: 29/08/2031 (based on 60-month shelf life)

Test Results

Parameter	Specification	Result	Method
Appearance	Colourless to white crystals	Conforms	Visual
Assay (as $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$)	99 – 103%	99.8%	Titrimetric
Melting Point	42 – 44°C	43.5°C	Capillary Tube
pH (5% solution in water, 25°C)	5.0 – 7.0	5.9	pH meter
Insoluble matter in water	Max 0.005%	0.003%	Gravimetric
Nitrite (NO_2)	Max 0.001%	0.0008%	Colorimetric
Chloride (Cl)	Max 0.005%	0.002%	Turbidimetric
Sulphate (SO_4)	Max 0.005%	0.004%	Turbidimetric
Barium (Ba)	Max 0.01%	0.005%	ICP-OES
Iron (Fe)	Max 0.0005%	0.0003%	ICP-OES
Lead (Pb)	Max 0.001%	0.0007%	ICP-OES
Magnesium (Mg)	Max 0.05%	0.03%	ICP-OES
Potassium (K)	Max 0.005%	0.002%	Flame Photometry
Sodium (Na)	Max 0.02%	0.015%	Flame Photometry
Strontium (Sr)	Max 0.05%	0.02%	ICP-OES

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Parameter	Specification	Result	Method
Density (typical, informational)	$\approx 1.86 \text{ g/cm}^3$	1.85 g/cm³	Hydrometer
Solubility in water (typical, informational)	121 g in 100 g of water	Conforms	Visual

Conclusion

All tested parameters conform to specification. **RESULT: PASS — APPROVED FOR RELEASE.**

Analyst

QC Manager

Authorised Signatory



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